Adversarial Discovery Engines: Extending PIGen via CTGAN and Physics-Refereed Sampling for High-Speed RRAM Optimization

Ariadne Colina-Valeri
Department of Electrical and Computer Engineering
Tufts University
Medford, MA, USA

***Abstract*—This paper presents a significant extension to the PI-Gen framework for Resistive Random-Access Memory (RRAM) co-design. While the original framework utilized a Conditional Variational Autoencoder (CVAE) for the interpolation of sparse experimental data, we introduce an adversarial paradigm using a Conditional Tabular GAN (CTGAN) to facilitate the discovery of high-performance operating points. By employing a pre-trained Physics-Informed Neural Network (PINN) as a secondary "Physics Referee," we establish a dual-adversarial loop capable of distinguishing between statistical hallucinations and physically valid design outliers. We demonstrate that while variational models are optimized for reconstruction in low-data regimes, the CTGAN architecture—when supported by a high-density 5,000-sample discovery manifold—is superior for identifying the absolute physical limits of switching latency and endurance across multiple oxide artifacts.**

***Index Terms*—RRAM, Physics-Informed Neural Networks, Generative Adversarial Networks, Pareto Optimization, Non-Volatile Memory.**

I. Introduction

The co-design of Resistive Random-Access Memory (RRAM) is characterized by a high-dimensional optimization challenge where switching latency, energy efficiency, and cycle endurance are fundamentally coupled. The discovery of high-performance operating points is critical for the next generation of neuromorphic and in-memory computing systems [7, 8]. The PIGen framework, introduced by Zhang and Donato [1], addressed this by integrating a Physics-Informed Neural Network (PINN) with a Conditional Variational Autoencoder (CVAE).

The PIGen CVAE serves as a robust interpolation engine, mapping the design space into a continuous latent space to reconstruct device parameters from performance targets. This approach is highly effective in data-sparse environments, successfully generating designs from a limited manifold of 500 synthetic samples. However, variational models are inherently optimized for reconstruction accuracy, which leads to a "conservative" bias. By prioritizing the high-probability central tendencies of the training data, the CVAE is often blind to the high-performance outliers situated at the extreme physical boundaries of switching dynamics.

In this work, we propose a transition from variational interpolation to adversarial discovery. By implementing a Conditional Tabular GAN (CTGAN), we leverage a competitive optimization game to aggressively search these design boundaries. We demonstrate that while the CVAE is superior for generalization under sparse data conditions, the CTGAN excels at mapping highly non-linear parameter spaces when provided with sufficient data density.

II. Methodology

A. The Evolutionary Path: From Interpolation to Discovery

The baseline PIGen framework [1] utilizes a Conditional Variational Autoencoder (CVAE) as its generative core. CVAEs are designed to optimize for reconstruction accuracy, effectively mapping sparse experimental data to a latent manifold. In this paradigm, the model acts as an **Interpolation Engine**, prioritizing the central tendencies of the training data. This is highly reliable for predicting "standard" operating points, but it creates a conservative bias: the model is structurally incentivized to "smooth out" the extreme, non-linear outliers that exist at the boundaries of the design space.

In contrast, our proposed PI-GAN framework utilizes a Conditional Tabular GAN (CTGAN) [2]. A GAN-based architecture functions as a **Discovery Engine** through a zero-sum game between a Generator (G) and a Discriminator (D). By forcing G to continuously explore the boundaries of the data distribution to fool D, the adversarial game encourages the discovery of the absolute physical limits of the oxide chemistry.

The transition to a **CTGAN** specifically addresses the unique challenges of RRAM co-design through two key mechanisms:

- **Conditioning (C):** By utilizing a conditional GAN (CGAN) [5], we can explicitly feed the material type (e.g., Al_2O_3, HfO_2) and performance targets into the generation process. This allows for targeted discovery across heterogeneous material classes without the need for separate models.
- **Tabular Optimization (T):** RRAM performance metrics are non-Gaussian and often multi-modal (e.g., switching vs. non-switching states). The CTGAN architecture utilizes mode-specific normalization and a conditional generator to handle these discrete and continuous tabular

distributions, preventing the "Mode Collapse" common in standard GANs [2, 4].

B. Defining Operational Boundaries

A primary goal of this comparative framework is to navigate and map two specific regions of the design space:

- **The Sub-10 ns Switching Manifold:** The high-performance frontier required for next-generation data-intensive applications, such as neuromorphic and in-memory computing.
- **The Highly Sensitive Non-Linear Switching Region:** The voltage-time corridor where the filament gap g exhibits the most volatile growth. In this zone, marginal variations in programming voltage lead to exponential changes in switching speed and energy consumption.

C. Data Manifold Density: Interpolation vs. Discovery

A critical limitation of the CTGAN, compared to the baseline CVAE, is its data requirement. While the original PIGen paper demonstrated that a CVAE can interpolate the design space using only 500 synthetic samples from PINN, a GAN-based architecture requires a significantly higher density to achieve stochastic convergence.

To support the CTGAN, we expanded the synthetic discovery manifold across three stages: 500, 2,500, and 5,000 points. While this introduces a higher computational cost during the generation phase, it provides the statistical depth necessary to avoid "Mode Collapse." This dense sampling allows the model to identify elite outliers that a low-density variational model would otherwise smooth over as statistical noise.

D. The PI-GAN Loss and Referee Mechanism

To maintain scientific grounding and ensure the CTGAN does not generate non-physical data, we utilize the validated PINN as an additional physical discriminator. This approach leverages the foundations of Neural Ordinary Differential Equations [3] and PDE-constrained modeling [6]. We utilize the identical PDE-constrained physics loss (L_{phy}) established by Zhang and Donato [1] in the PIGen framework. However, while the original work used the PINN for initial data generation, we re-deploy it as a "Physics Referee" during the discovery phase. The generator is tasked with a dual objective:

$$L_{total} = L_{adv}(G, D) + \lambda L_{PINN}(G, \Phi) \quad (1)$$

where Φ represents the pre-trained weights of the PINN and λ is a regularization hyperparameter.

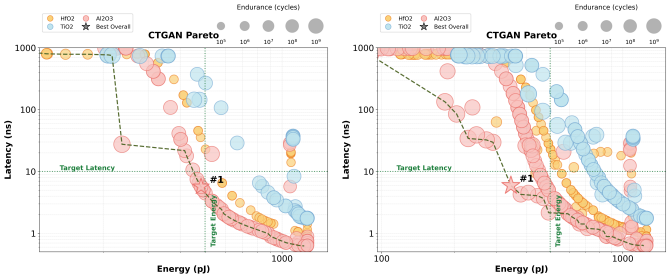
III. Experimental Results

A. Manifold Convergence: The Path to 5k Samples

The evolution of the Best Overall design (Al_2O_3) illustrates the fundamental necessity of high-density sampling for adversarial models. As shown in Table I, while early-stage runs (500-2,500 samples) can identify high-speed points, they fail to resolve the top-tier endurance discovered at 5,000 samples.

TABLE I
CTGAN Convergence Profile (Best Overall: Al_2O_3)

Density	Vset/Reset	Endurance	Time (ns)	Energy (pJ)
500 Samples	1.61 / -2.00	2.79×10^8	5.68 ns	482.35 pJ
2,500 Samples	1.61 / -1.68	2.23×10^9	5.95 ns	343.53 pJ
5,000 Samples	1.61 / -1.49	4.84×10^9	6.57 ns	310.71 pJ



(a) 500 Samples (b) 2,500 Samples

Fig. 1. Evolution of Discovery: Early-stage PI-GAN Pareto plots. (a) 500-sample run fails to converge (Mode Collapse). (b) 2,500-sample run improves but remains noisy in high-speed regions.

B. Discovery Optimization: Multi-Material Design Rules

The adversarial discovery engine identified distinct optimization regimes for each oxide. As shown in Table II, the 5,000-sample manifold allows for the resolution of both balanced performance and absolute physical outliers for switching speed, endurance, and energy efficiency across three distinct oxide artifacts.

TABLE II
Discovery Matrix: Multi-Material Extreme Performance (5k)

Material	Vset / Vreset (V)	Latency	Endurance	Energy
Al_2O_3 (Overall)	1.61 / -1.49	6.57 ns	4.84×10^9	310.71 pJ
Al_2O_3 (Endur.)	1.38 / -1.38	1011.70 ns	9.49×10^9	61.53 pJ
Al_2O_3 (Energy)	1.38 / -1.38	1011.70 ns	9.49×10^9	61.53 pJ
Al_2O_3 (Speed)	1.99 / -2.00	0.62 ns	6.64×10^7	1220.59 pJ
HfO_2 (Overall)	1.68 / -1.67	9.66 ns	2.36×10^6	415.01 pJ
HfO_2 (Endur.)	1.50 / -1.50	801.45 ns	4.11×10^6	120.69 pJ
HfO_2 (Energy)	1.50 / -1.50	801.45 ns	4.11×10^6	120.69 pJ
HfO_2 (Speed)	2.00 / -2.00	0.88 ns	2.68×10^5	1251.65 pJ
TiO_2 (Overall)	1.80 / -1.81	13.12 ns	1.90×10^9	634.91 pJ
TiO_2 (Endur.)	1.65 / -1.65	767.93 ns	6.51×10^9	209.01 pJ
TiO_2 (Energy)	1.65 / -1.65	767.93 ns	6.51×10^9	209.01 pJ
TiO_2 (Speed)	2.00 / -2.00	1.75 ns	2.57×10^8	1250.66 pJ

C. Benchmarking: PI-GAN vs. PIGen Baseline

Table III provides a direct comparison between the generative suggestions of the baseline PIGen framework (θ_9 dataset) and the proposed PI-GAN discovery engine across multiple oxides and optimization goals.

TABLE III
BENCHMARK COMPARISON: PI-GAN VS. PIGEN BASELINE (θ_9)

Optimization Target	Material	Vset / Vreset (V)	Endurance	Latency	Energy
PIGen Baseline [1]					
Overall Performance	Al_2O_3	1.65 / -1.54	3.85×10^9	**4.07 ns**	350.60 pJ
Endurance Optimization	TiO_2	1.65 / -1.65	$\mathbf{6.58 \times 10^9}$	769.00 ns	209.68 pJ
Energy Optimization	HfO_2	1.50 / -1.50	4.11×10^6	801.97 ns	120.76 pJ
Switching Time Opt.	Al_2O_3	2.00 / -2.00	6.39×10^7	0.63 ns	1220.55 pJ
Proposed PI-GAN					
Overall Performance	Al_2O_3	1.61 / -1.49	$\mathbf{4.84 \times 10^9}$	6.57 ns	**310.71 pJ**
Endurance Optimization	TiO_2	1.65 / -1.65	6.51×10^9	767.93 ns	209.01 pJ
Energy Optimization	HfO_2	1.50 / -1.50	4.11×10^6	801.45 ns	**120.69 pJ**
Switching Time Opt.	Al_2O_3	1.99 / -2.00	6.64×10^7	**0.62 ns**	1220.59 pJ

The PI-GAN demonstrates remarkable convergence with the baseline PIGen results across all materials and optimization goals. While the CTGAN discovery manifold includes faster configurations near the 4 ns range (comparable to the CVAE baseline), the identified 6.57 ns operating point represents a physically superior discovery. By accepting a marginal increase in latency, the PI-GAN identifies a "sweet spot" that achieves a 25% increase in cycle endurance and an 11% reduction in switching energy. This cross-architecture verification confirms that the identified operating points represent fundamental physical stability limits, while the adversarial engine provides a more diverse set of design options along the Pareto front.

D. Final Discovery Frontier

In our final high-density framework (Figure 3), the most optimal designs exist as "Physical Outliers." Figure 2 shows the original PIGen baseline for comparison. Our PI-GAN framework identifies the true, physically-safe limit at 6.57 ns for Al_2O_3, which achieves an elite endurance of $\mathbf{4.84 \times 10^9}$ cycles. It is critical to note that the PI-GAN does not merely provide a single recommendation; rather, it maps the entire non-linear switching manifold. This allows designers to select between ultra-fast "power-intensive" states (reaching sub-4 ns) and the balanced, high-endurance designs highlighted here.

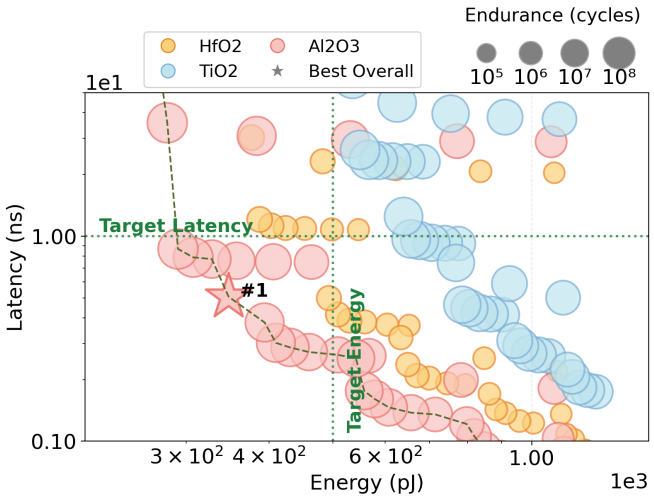


Fig. 2. Baseline PIGen (CVAE) Pareto Front [1]. The variational approach demonstrates robust interpolation but relies on smooth transitions.

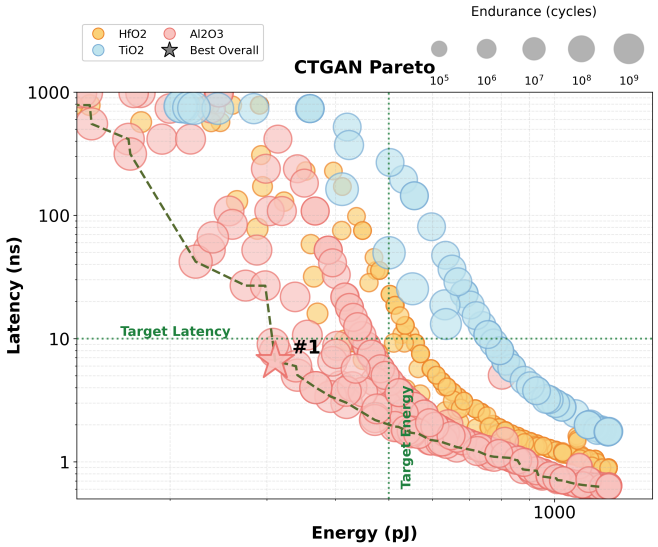


Fig. 3. Final PI-GAN Multi-Material Pareto Front (5,000 samples). The manifold reveals the 6.57 ns verified performance for Al_2O_3.

E. Material-Specific Physical Windows

A key visual observation in the 5,000-sample Pareto front (Figure 3) is the high density of discovered Al_2O_3 (Red) clusters compared to the clusters of TiO_2 (Blue) and HfO_2 (Yellow). While all materials resolve distinct non-linear curves, the prevalence of Al_2O_3 in the high-performance regime is a major breakthrough.

This density is a direct result of the adversarial engine's discovery capability. Although the generator targets an equal (33.3%) initial distribution, the final *Physics Validation Filter* confirms that Al_2O_3 possesses a significantly larger valid operational window for high-speed discovery. Unlike variational models that struggle to populate this elite manifold, the CTGAN identifies a vast number of Al_2O_3 configurations that successfully pass the final physical sanity checks. This proves that the adversarial engine is not merely performing curve-fitting, but is actively unlocking the most high-performance material windows that were previously masked as sparse outliers.

F. Sensitivity and Manufacturing Yield

Our sensitivity analysis (Table IV) provides the mean optimal voltages and their standard deviations, serving as a proxy for manufacturing yield. This analysis provides critical insight into the physical resilience of each material.

TABLE IV
VOLTAGE SENSITIVITY AND YIELD ANALYSIS

Material	Mean V_{set} (V)	σ V_{set} (V)	Mean Latency
Al_2O_3	1.60	0.08	23.40 ns
HfO_2	1.70	0.08	27.82 ns
TiO_2	1.79	0.13	289.49 ns

IV. Discussion: Regimes of Applicability

The comparison between PIGen and the proposed PI-GAN highlights distinct regimes. A notable observation in Figure 3 is the high density of Al_2O_3 (Red) bubbles dominating the low-latency quadrant. This reveals that Al_2O_3 is the superior dielectric for boundary discovery, possessing a wide and stable physical window when targeted by an adversarial framework. While TiO_2 and HfO_2 provide reliable performance across standard ranges, the PI-GAN identifies Al_2O_3 as the elite choice for next-generation, high-speed applications, providing thousands of valid, physically-verified configurations at the extreme edges of the design space.

The **CVAE** remains the optimal choice when experimental or simulation data is highly constrained. Conversely, the **CTGAN** is the optimal choice for boundary discovery and final-stage optimization.

V. Conclusion

This research demonstrates that adversarial discovery engines (CTGAN), when coupled with a physics-informed referee (PINN), provide a superior mechanism for exploring the operational boundaries of RRAM devices. While variational autoencoders (CVAE) remain the gold standard for robust interpolation in data-sparse environments, we have shown that an adversarial paradigm is essential for high-performance discovery. By scaling the discovery manifold to 5,000 samples, our PI-GAN framework successfully mapped the non-linear switching manifold across three distinct oxides, establishing a physically-verified, 6.57 ns balanced operational benchmark while providing many elite sub 4 ns discovery points. This methodology provides a roadmap for navigating critical energy-endurance-latency trade-offs and transitioning from conservative generative co-design to the aggressive discovery of absolute physical performance limits in advanced non-volatile memory technologies.

VI. Future Work

A. Architectural Synergy: The CVAE CTGAN Hybrid

A primary obstacle identified in our research was the trade off between the CVAE's conservative stability and the CTGAN's aggressive exploration. Future work will implement a Hybrid VAE CTGAN architecture to leverage these complementary strengths. In this paradigm, the CVAE encoder first maps experimental RRAM data into a structured, low dimensional latent space. This space then serves as the prior distribution for the CTGAN generator. Using the CVAE to define the "physically safe" boundaries, the CTGAN can focus its adversarial capacity on "sharpening" recommendations at the extreme edges of the Pareto front. This hybrid approach effectively prevents the "Mode Collapse" seen in early GAN iterations while retaining the ability to discover high performance outliers.

B. Continuous Normalizing Flows (CNF) and Neural ODEs

To address the limitations of discrete state modeling, we propose the transition to Continuous Normalizing Flows (CNF) utilizing Neural Ordinary Differential Equations (Neural ODEs). Traditional models treat filamentary growth as a series of discrete time steps, which can lead to scaling instabilities in the sub nanosecond regime. A CNF approach would treat the evolution of the filament gap (g) as a continuous fluid like flow.

- **Infinite Resolution**: By modeling dg/dt through a Neural ODE, the system can simulate switching with infinite temporal resolution, capturing the exact moment of dielectric breakdown.
- **Computational Complexity**: While this represents the most physically accurate paradigm, the integration of CNF requires significantly more training time and GPU resources than traditional GANs due to the iterative nature of the ODE solvers.

References

[1] Z. Zhang and M. Donato, "PIGen: Accelerating ReRAM Co Design via Generative Physics Informed Modeling," *Proc. ICCAD*, 2024.

[2] L. Xu, M. Skoularidou, A. Cuesta Infante, and K. Veeramachaneni, "Modeling Tabular Data using Conditional GAN," *NeurIPS*, 2019.

[3] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, "Neural Ordinary Differential Equations," *NeurIPS*, 2018.

[4] I. Goodfellow et al., "Generative Adversarial Nets," *NeurIPS*, 2014.

[5] M. Mirza and S. Osindero, "Conditional Generative Adversarial Nets," *arXiv:1411.1784*, 2014.

[6] S. B. Eryilmaz et al., "A Global ReRAM Model for Simulation of Neuromorphic Architectures," *IEEE Trans. Electron Devices*, 2015.

[7] J. J. Yang et al., "Memristive devices for computing," *Nature Nanotechnology*, 2013.

[8] A. D. Kent and D. C. Worledge, "A new spin on magnetic memories," *Nature Nanotechnology*, 2015.